2 A DISRUPTIVE VR PLATFORM FOR REAL ESTATE

Designing and constructing a building is not an easy task. Lack of proper research and lucid documentation can cause confusion and lead to errors/defects in work or a compromise in quality. Rework further increases the cost.

Realising the potential of virtual reality (VR) in mitigating these issues, husband-wife duo Tithi G. Tewari and Gautam Tewari founded SmartVizX in 2015 with the aim to maximise business outcomes by implementing virtualisation solutions. Target consumer segments for its VR technology consist of architects, interior designers, building product manufacturers and suppliers, engineering service consultants, developers, owners and operators.

The company raised its first funding from Indian Angel Network (IAN) and Stanford Angels & Entrepreneurs India (SA&E India). Its team comprises professionals from fields of architecture and gaming technology. As the first line of products, the startup launched several VR tools (like Arch Viz, 360-Viz and Vi-Viz) and services for business use.

Arch Viz software allows visualisation of real estate projects, and 360-Viz lets users experience any existing property without actually visiting the place. Web-based platform Vi-Viz enables users to interact with properties and products, and obtain information about every aspect. Most of these tools have also been launched on mobile platforms. This has led to the creation of Trezi in 2018, a VR-based platform for architecture and interior design collaboration.

Selection and conversion of architectural models, BIM objects or product catalogues to VR is a smooth process. Users just need to launch their 3D architecture/interior design or building product model into the workspace. The platform enables users to instantly experience their work at full scale and colour, which can be presented to clients, colleagues and stakeholders for review.

Available editing features, material libraries and product catalogues in the workspace enable exploring and evaluating suitable design options. The refined project can finally be shared in a variety of output formats.

The platform's development was further fuelled after it secured funding from IAN Fund and YourNest in 2018. Its latest version 1.8 is compatible with architecture design software like Sketchup, Rhino and Autodesk's Revit. Architects and designers can experience a true-to-scale presentation of their work remotely and make edits in real time. Exterior rendering mode also comes with improved quality.

Introducing the solution in the workflow bridges the communication gap that exists in the architecture-engineering-construction (AEC) industry. This results in better awareness and perception of design elements as well as improved collaboration, leading to improved choices.

By resolving issues early in the project lifecycle, the actual constructed project has negligible defects for the first time itself. The benefit is decreased material wastage resulting from physical models and mockups. Expenses are lower, projects have fast turnaround times and overall client satisfaction is improved, which make the business more profitable.

A fourteen-day free trial is available to try out the solution. The startup has worked with Rockworth, ASF Group, Oberoi Realty, Quikr, Edifice and Godrej, among others. Many international clients are already using Trezi. The company is currently expanding its focus from the Indian market to global markets.

—Ayushee Sharma

3 LIDAR FOR ENVIRONMENT SENSING IN ROBOTAXI

True autonomous driving requires human-like advanced sensing of the environment, for driving with little or no human input. Safety of passengers and passersby is the foremost priority in autonomous driving. To achieve the same, RoboSense delivers comprehensive intelligent environment perception lidar solutions, including chip, FPGA, lidar hardware and artificial intelligence (AI) algorithms. The lidar solution, RS-Fusion-P5, has been used in various application scenarios, such as in self-driving logistics vehicles, buses, passenger cars and so on. This ensures a safer and more-efficient autonomous driving for RoboTaxi.

RS-Ruby and RS-Bpearl

RoboSense RS-Fusion-P5 uses a combination of RS-Ruby and RS-Bpearl to ensure a sensing range within a radius of more